

Abel Rajesh

Citywest, Dublin | [Email](#) | +353 89 495 8035 | [LinkedIn](#)

Personal Profile

First Class Honours Physics with Data Analytics graduate and MSc in Advanced AI student. Experienced in Python data engineering, SQL, Power BI and RAG architectures. Demonstrated cross-functional collaboration and analytical problem-solving in R&D environments. Eager to leverage generative AI, workflow automation and responsible AI governance to support practical solution delivery, ensuring seamless integration through comprehensive documentation and clear stakeholder communication.

Technical Skills

Data Analytics & Visualisation: Power BI, Dashboard Development, KPI Reporting, SQL, Data Cleaning, Statistical Analysis, Matplotlib

Software Engineering: Python (Pandas, NumPy, Scikit-learn, TensorFlow), REST APIs, Linux, Git/GitHub, Docker, Unit Testing, System Architecture

Data & AI Systems: Automated ETL Data Pipelines, MLOps, CI/CD, Model Evaluation, Remote Hardware Protocols (SSH/VNC), RAG Architecture, Agentic Workflow Patterns, Responsible AI Governance

Education

University College Dublin
MSc in Advanced AI

2026 - 2027

Relevant Modules: Responsible AI, ML System Deployment, Agentic AI, AI for Time Series, Advanced Generative AI & LLMs

Dublin City University
BSc in Physics with Data Analytics: First Class Honours (1.1)

2022 - 2026

Relevant Modules: High Performance Computing, Linear Algebra, Numerical Methods, Advanced Programming, Machine Learning, Data Analytics & Data Mining

Selected Projects:

Stock-Market Sector Growth & Time-Series Forecasting: Engineered automated data pipelines integrating the Finnhub API to map stock tickers across the NASDAQ, S&P 500 and NYSE into market sectors. Developed time-series forecasting models (Linear Regression, ARIMA, LSTM) to predict sector growth trends and price movements. Evaluated out-of-sample model performance using MAE, RMSE, and R^2 statistical metrics to optimise data processing workflows for large-scale financial analysis.

Modular ML & GenAI Evaluation: Developed a modular classification and evaluation harness applying scikit-learn to map multi-class inputs into structured feature spaces. Evaluated Decision Trees, SVM and KNN algorithms using confusion matrices and hypothesis testing to identify risk vectors and out-of-distribution errors. Integrated retrieval-augmented generation (RAG) structures and agentic workflow patterns to test automated evaluation harness designs for LLM quality.

Work History

Tomra Sorting Limited, Citywest
R&D Engineer - Internship

Feb 2025 - Sep 2025; May 2026 - Oct 2026

- Authored technical user documentation and Standard Operating Procedures (SOPs) for a local image-scanning tool, enabling the offline testing and evaluation of classification models
- Transformed live sensor and system data into operational insights for cross-functional teams, supporting continuous process improvement and quality control
- Engineered production Python data pipelines and real-time monitoring tools to process live, high-frequency sensor streams, detecting anomalies and optimising data execution speed
- Implemented unsupervised DBSCAN clustering and machine vision algorithms to segment data objects in high-throughput operational environments
- Executed remote deployments and automated integration testing via Linux, VNC and SSH protocols across operational hardware to ensure system stability
- Curated and validated high-volume datasets for deep learning models, identifying edge cases and boosting classification accuracy to 97%

Uniphar Group, Citywest
Warehouse Operative

July 2022 - Feb 2025

- Coordinated cross-functional workflow improvements to ensure zero-defect compliance across physical and digital inventory systems

Achievements & Languages

Irish International Taekwondo: 3x Medalist (South Dublin Representative)

Music: Grade 4 Guitar - Trinity College London

Licence: Full Irish Driver's Licence

Spanish/Irish: Intermediate (Secondary School)

Malayalam: Fluent

Interests

Active in fitness (gym 4x weekly); Teach guitar to children; Practice Taekwondo; Swim weekly

References

References available upon request